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WITHOUT MUTEX

```
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define INCREMENTS 100000

int counter = 0;

void* increment(void* arg) {
    for(int i = 0; i < INCREMENTS; i++) {
        counter++;
    }
    return NULL;
}

int main() {

    pthread_t threads[THREADS];

    // Create threads
    for(int i = 0; i < THREADS; i++) {
        pthread_create(&threads[i], NULL, increment, NULL);
    }

    // Wait for threads to finish
    for(int i = 0; i < THREADS; i++) {
        pthread_join(threads[i], NULL);
    }

    printf("Final Counter Value (Without Mutex) = %d\n", counter);

    return 0;
}
```

WITH MUTEX:

```
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
```

```
#define INCREMENTS 100000

int counter = 0;

pthread_mutex_t lock;

void* increment(void* arg) {

    for(int i = 0; i < INCREMENTS; i++) {

        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);

    }

    return NULL;
}

int main() {

    pthread_t threads[THREADS];

    pthread_mutex_init(&lock, NULL);

    for(int i = 0; i < THREADS; i++) {
        pthread_create(&threads[i], NULL, increment, NULL);
    }

    for(int i = 0; i < THREADS; i++) {
        pthread_join(threads[i], NULL);
    }

    pthread_mutex_destroy(&lock);

    printf("Final Counter Value = %d\n", counter);

    return 0;
}
```

OUTPUT:

```
PS C:\Users\prerana\OneDrive\Desktop\pthread_program> gcc race_condition.c -o race_condition -lpthread
PS C:\Users\prerana\OneDrive\Desktop\pthread_program> ./race_condition.exe
Final Counter Value (Without Mutex) = 197964
PS C:\Users\prerana\OneDrive\Desktop\pthread_program> gcc mutex_counter.c -o mutex_counter -lpthread
PS C:\Users\prerana\OneDrive\Desktop\pthread_program> ./mutex_counter.exe
Final Counter Value = 400000
PS C:\Users\prerana\OneDrive\Desktop\pthread_program> █
```